

INTEGRATION GUIDE

Turing Service API

Multi-tenant voice calling platform — complete API reference for integration partners, frontend developers, and platform operators.

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Base URL: `https://<your-domain>:8005`

Endpoints: 68

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TABLE OF CONTENTS

- [01. Overview & Architecture](#)
- [02. Quick Start](#)
- [03. Authentication](#)
- [04. Error Handling](#)
- [05. Rate Limiting](#)
- [06. Health & Readiness](#)
- [07. Client Registration](#)
- [08. Portal & Claim Links](#)
- [09. Admin Console](#)
- [10. Self-Service \(Me\)](#)
- [11. Calls](#)
- [12. Batches](#)
- [13. Analytics](#)
- [14. Agents](#)
- [15. Phone Numbers](#)
- [16. SQL Builder Agent \(Paused\)](#)
- [17. Webhooks](#)
- [18. Schema Reference](#)
- [19. Environment Variables](#)
- [20. Changelog](#)

01. Overview & Architecture

Turing Service is a **multi-tenant voice calling orchestration platform** built with FastAPI. It acts as a management and analytics layer on top of the Bolna voice engine, providing:

- **Multi-tenant isolation** — each client (tenant) gets their own API keys, agent configurations, phone number assignments, and call data
- **Batch call management** — upload CSV or JSON recipient lists, schedule calls, monitor execution
- **Real-time analytics** — call volume, duration, cost, and outcome breakdowns with time-series support
- **LLM-powered call analysis** — automatic classification of call outcomes (booking, escalation, follow-up, etc.)
- **Admin console** — full lifecycle management of clients, agents, phone numbers, and drift detection
- **SQL Builder Agent** — natural-language-to-SQL query interface for analytics

Technology Stack

COMPONENT	TECHNOLOGY
Framework	FastAPI (Python 3.11+)
Database	PostgreSQL (asyncpg via SQLAlchemy 2.0)
Cache / Rate Limiting	Redis (with in-memory fallback)
Voice Engine	Bolna AI (<code>api.bolna.ai</code>)
LLM Providers	Anthropic, OpenAI, Google (configurable)
Migrations	Alembic

Authentication Zones

The API is divided into four authentication zones, enforced by middleware:

ZONE	PATHS	AUTH METHOD	USE CASE
Open	<code>/health/*</code> , <code>/v1/register</code> , <code>/v1/portal/*</code> , <code>/v1/claim/*</code>	None	Health checks, onboarding
Admin	<code>/v1/admin/*</code>	X-Admin-Key header	Platform operators
Tenant	<code>/v1/me/*</code> , <code>/v1/calls/*</code> , <code>/v1/batches/*</code> , etc.	X-API-Key header	Integration partners

ZONE	PATHS	AUTH METHOD	USE CASE
IP Allowlist	/webhooks/voice	Source IP verification	Voice engine callbacks

02. Quick Start

Get from zero to your first API call in three steps.

Step 1 — Register

```
curl -X POST https://your-domain:8005/v1/register \
  -H "Content-Type: application/json" \
  -d '{"name": "Acme Health", "contact_email": "dev@acme.com}"'
```

Step 2 — Get Your API Key

After an admin approves your registration, you'll receive an API key via the claim link (or directly from the admin). The key looks like `tsk_abc123...`.

Step 3 — Make Your First Call

```
curl https://your-domain:8005/v1/calls \
  -H "X-API-Key: tsk_your_api_key_here" \
  -H "Content-Type: application/json"

# Response: list of your calls (empty initially)
```

Step 4 — Create a Batch

```
curl -X POST https://your-domain:8005/v1/batches \
  -H "X-API-Key: tsk_your_api_key_here" \
  -H "Content-Type: application/json" \
  -d '{
    "agent_id": "your-agent-id",
    "recipients": [
      {"recipient_phone_number": "+919876543210", "user_data": {"name": "John"}}
    ]
  }'
```

03. Authentication

Tenant API Key (X-API-Key)

All tenant-scoped endpoints require the `X-API-Key` header. Keys are issued per-client and can have multiple active keys with optional labels.

```
GET /v1/calls HTTP/1.1
Host: your-domain:8005
X-API-Key: tsk_abc123def456...
```

Key Caching: API keys are cached for `API_KEY_CACHE_TTL_SECONDS` (default 60s) after first lookup. Key revocation takes effect after cache expiry.

Admin Key (X-Admin-Key)

All `/v1/admin/*` endpoints require the `X-Admin-Key` header. This is a single shared secret set via the `ADMIN_API_KEY` environment variable.

```
GET /v1/admin/clients HTTP/1.1
Host: your-domain:8005
X-Admin-Key: your-admin-secret
```

Webhook IP Allowlist

The `/webhooks/voice` endpoint restricts access by source IP. Configure allowed IPs via `VOICE_WEBHOOK_ALLOWED_IPS` (comma-separated). An empty value allows all IPs.

Error Responses

STATUS	CONDITION
401 Unauthorized	Missing or invalid API key
403 Forbidden	Client is suspended or inactive

04. Error Handling

All errors follow a consistent JSON envelope:

```
{
  "error": "validation_error",
  "message": "Human-readable description",
  "detail": { ... },
  "request_id": "550e8400-e29b-41d4-a716-446655440000"
}
```

Error Codes

HTTP STATUS	ERROR CODE	DESCRIPTION
400	http_error	Bad request — check the detail field
401	http_error	Authentication required or invalid credentials
403	http_error	Client suspended or insufficient permissions
404	http_error	Resource not found
422	validation_error	Request body failed Pydantic validation
429	http_error	Rate limit exceeded
500	internal_error	Unexpected server error
502	voice_engine_error	Upstream voice engine failure
502	llm_error	LLM provider failure (analysis / SQL agent)

Request ID: Every response includes an X-Request-ID header. Include this when reporting issues for faster debugging.

05. Rate Limiting

Rate limiting is enforced at the service layer using Redis (with an in-memory fallback when Redis is unavailable).

ENDPOINT	LIMIT	WINDOW
POST /v1/register	REGISTER_RATE_LIMIT_PER_HOUR (default: 10)	1 hour, per IP

When a limit is exceeded, the API returns `429 Too Many Requests`. Implement exponential backoff in your client.

06. Health & Readiness

Use these endpoints for load balancer health checks and deployment readiness probes.

METHOD	PATH	AUTH	DESCRIPTION
GET	/health	Open	Basic liveness probe
GET	/health/ready	Open	Readiness probe (checks DB connectivity)

Response — GET /health

```
{
  "status": "ok",
  "service": "turing_service",
  "version": "1.0.0",
  "environment": "production"
}
```

07. Client Registration

New clients self-register and wait for admin approval before receiving an API key.

METHOD	PATH	AUTH	DESCRIPTION
POST	/v1/register	Open	Register a new client

Request Body

RegisterRequest			
FIELD	TYPE	CONSTRAINTS	DESCRIPTION
name	string	required 1–128 chars	Organization name
contact_email	string null	optional max 256	Contact email address

Response — 200 OK

```
{
  "status": "pending",
  "message": "Registration submitted. You will be notified once approved."
}
```


08. Portal & Claim Links

The portal enables clients to look up their account, and claim links provide a one-time secure mechanism for distributing API keys.

METHOD	PATH	AUTH	DESCRIPTION
POST	/v1/portal/lookup	Open	Look up client by name or email
GET	/v1/claim/{token}	Open	Preview claim link (non-destructive)
POST	/v1/claim/{token}	Open	Burn claim link and receive API key

Claim Flow

1. Admin approves a client → system generates a one-time claim URL
2. Client visits `GET /v1/claim/{token}` to preview (see org name, expiry)
3. Client calls `POST /v1/claim/{token}` to consume the token and receive their API key

One-time use: Claim tokens expire after `CLAIM_LINK_TTL_HOURS` (default 24h) and are destroyed on first successful consumption. Store the returned API key securely.

Response — `POST /v1/claim/{token}`

```
{
  "client_name": "Acme Health",
  "api_key": "tsk_live_a1b2c3d4e5f6..."
}
```

09. Admin Console

The admin console provides full lifecycle management for all platform resources. All endpoints require `X-Admin-Key`.

Client Management 9 endpoints

METHOD	PATH	DESCRIPTION
POST	/v1/admin/clients	Create a new client
GET	/v1/admin/clients	List all clients (filterable by status)
GET	/v1/admin/clients/{client_id}	Get client details
PATCH	/v1/admin/clients/{client_id}	Update client name or email
DELETE	/v1/admin/clients/{client_id}	Delete a client
POST	/v1/admin/clients/{client_id}/approve	Approve a pending client (issues API key)
POST	/v1/admin/clients/{client_id}/reject	Reject a pending client
POST	/v1/admin/clients/{client_id}/suspend	Suspend an active client
POST	/v1/admin/clients/{client_id}/reactivate	Reactivate a suspended client

API Key Management 3 endpoints

METHOD	PATH	DESCRIPTION
GET	/v1/admin/clients/{client_id}/keys	List client's API keys
POST	/v1/admin/clients/{client_id}/keys	Issue a new API key
DELETE	/v1/admin/clients/{client_id}/keys/{key_id}	Revoke an API key

Client Configuration 2 endpoints

METHOD	PATH	DESCRIPTION
GET	/v1/admin/clients/{client_id}/config	Get client configuration
PUT	/v1/admin/clients/{client_id}/config	Update client configuration

Configuration Fields

ClientConfigUpdate		
FIELD	TYPE	DESCRIPTION
default_from_number	string null	Default caller ID for outbound calls
webhook_url	string null	URL to receive call outcome webhooks
webhook_secret	string null	HMAC secret for webhook signature verification
visible_fields	object null	Fields visible in the client portal
settings	object null	Custom settings JSON
analysis_llm_provider	string null	LLM provider for call analysis (anthropic , openai)
analysis_llm_model	string null	Model override (e.g. gpt-4o)
analysis_prompt_hint	string null	Custom prompt hint appended to analysis
analysis_llm_api_key	string null	Client's own LLM API key (encrypted at rest)

Agent Management 7 endpoints

METHOD	PATH	DESCRIPTION
GET	/v1/admin/agents	List all agents in the catalog
GET	/v1/admin/agents/{agent_id}/variables	Get agent prompt variables
GET	/v1/admin/clients/{client_id}/agents	List agents assigned to a client

METHOD	PATH	DESCRIPTION
PUT	/v1/admin/clients/{client_id}/agents	Set which agents a client can use
PATCH	/v1/admin/clients/{client_id}/agents/{voice_agent_id}	Update per-client agent config (display name, variable overrides)
GET	/v1/admin/clients/{client_id}/drift	List drift events for a client
POST	/v1/admin/clients/{client_id}/drift/{event_id}/acknowledge	Acknowledge a drift event

Sync Operations 2 endpoints

METHOD	PATH	DESCRIPTION
POST	/v1/admin/agents/sync	Sync agent catalog from voice engine
POST	/v1/admin/phone-numbers/sync	Sync phone number catalog from voice engine

Voice Engine & Phone Numbers 4 endpoints

METHOD	PATH	DESCRIPTION
GET	/v1/admin/voice-engine/status	Check voice engine connectivity
GET	/v1/admin/phone-numbers	List all phone numbers in catalog
GET	/v1/admin/clients/{client_id}/phone-numbers	List phone numbers assigned to a client
PUT	/v1/admin/clients/{client_id}/phone-numbers	Assign phone numbers to a client

Admin Calls & Analytics 8 endpoints

METHOD	PATH	DESCRIPTION
GET	/v1/admin/clients/{client_id}/calls	List calls for a client
GET	/v1/admin/clients/{client_id}/calls/{call_id}	Get call detail (transcript, analysis)

METHOD	PATH	DESCRIPTION
POST	/v1/admin/clients/{client_id}/calls/{call_id}/analyze	Trigger LLM analysis for a call
GET	/v1/admin/clients/{client_id}/batches	List batches for a client
GET	/v1/admin/clients/{client_id}/analytics/overview	Analytics overview for a client
GET	/v1/admin/clients/{client_id}/analytics/by-agent	Analytics grouped by agent
GET	/v1/admin/clients/{client_id}/analytics/by-batch	Analytics grouped by batch
GET	/v1/admin/clients/{client_id}/analytics/timeseries	Time-series analytics data

10. Self-Service (Me)

Tenant-facing endpoints for clients to manage their own account. Requires `X-API-Key` .

METHOD	PATH	DESCRIPTION
GET	/v1/me	Get own client profile
GET	/v1/me/config	Get own configuration
GET	/v1/me/keys	List own API keys
POST	/v1/me/keys	Issue a new API key for self
DELETE	/v1/me/keys/{key_id}	Revoke own API key

Response — GET /v1/me

```
{
  "client_id": "550e8400-e29b-41d4-a716-446655440000",
  "name": "Acme Health",
  "slug": "acme-health",
  "contact_email": "dev@acme.com",
  "status": "active",
  "created_at": "2025-01-15T10:30:00Z",
  "approved_at": "2025-01-15T11:00:00Z",
  "active_key_count": 2
}
```

11. Calls

Manage and monitor individual voice calls. Requires `X-API-Key` .

METHOD	PATH	DESCRIPTION
GET	/v1/calls	List calls (paginated, filterable)
GET	/v1/calls/{execution_id}	Get call detail (transcript, extracted data, analysis)
POST	/v1/calls/{execution_id}/stop	Stop an in-progress call
POST	/v1/calls/{execution_id}/analyze	Trigger LLM analysis for a call

Response — GET /v1/calls (paginated)

```
{
  "items": [
    {
      "call_id": "exec_abc123",
      "agent_id": "agent-001",
      "batch_id": "550e8400-...",
      "contact_number": "+919876543210",
      "from_number": "+911234567890",
      "status": "completed",
      "duration": 45.2,
      "cost": 0.12,
      "hangup_reason": "customer_ended",
      "recording_url": "https://...",
      "created_at": "2025-01-20T14:30:00Z",
      "analysis": {
        "outcome": "booking",
        "summary": "Patient confirmed appointment for Jan 25",
        "confidence": 0.92
      }
    }
  ],
  "total": 150,
  "page": 1,
  "page_size": 20,
  "pages": 8
}
```

Call Analysis

The `POST /v1/calls/{execution_id}/analyze` endpoint triggers LLM-powered analysis of a call's transcript.

Outcome Categories

OUTCOME	DESCRIPTION
booking	Appointment or action was confirmed
escalation	Call was escalated to a human
not_interested	Recipient declined or hung up
follow_up	Requires a follow-up call
no_output	No meaningful conversation occurred
not_reached	Call did not connect
other	Doesn't fit other categories

12. Batches

Create and manage batch calling campaigns. Requires `X-API-Key`.

METHOD	PATH	DESCRIPTION
POST	/v1/batches	Create a batch from JSON recipients
POST	/v1/batches/upload	Create a batch from CSV upload
POST	/v1/batches/{batch_id}/schedule	Schedule a batch for execution
GET	/v1/batches	List all batches (paginated)
GET	/v1/batches/by-agent/{agent_id}	List batches for a specific agent
GET	/v1/batches/{batch_id}	Get batch details
GET	/v1/batches/{batch_id}/executions	List individual call executions in a batch
GET	/v1/batches/{batch_id}/metrics	Get batch metrics (completion, cost, duration)
POST	/v1/batches/{batch_id}/stop	Stop all pending calls in a batch
DELETE	/v1/batches/{batch_id}	Delete a batch

Create Batch — Request Body

CreateBatchRequest			
FIELD	TYPE	CONSTRAINTS	DESCRIPTION
agent_id	string	required	Voice agent ID (must be enabled for your tenant)
recipients	array[object]	required min 1	List of recipients with phone numbers and user data
from_phone_numbers	array[string] null	optional	Caller IDs to rotate through
retry_config	RetryConfig null	optional	Retry policy for failed calls

FIELD	TYPE	CONSTRAINTS	DESCRIPTION
<code>webhook_url</code>	<code>string null</code>	optional	Override webhook URL for this batch

Retry Configuration

RetryConfig		
FIELD	TYPE	DESCRIPTION
<code>enabled</code>	<code>boolean</code>	Enable/disable retries
<code>max_retries</code>	<code>integer</code>	Maximum retry attempts
<code>retry_on_statuses</code>	<code>array[string]</code>	Statuses that trigger a retry
<code>retry_on_voicemail</code>	<code>boolean</code>	Retry if call hits voicemail
<code>retry_intervals_minutes</code>	<code>array[integer]</code>	Delays between retries (min 15 min each)

Example — Create & Schedule a Batch

```
# 1. Create
curl -X POST https://your-domain:8005/v1/batches \
  -H "X-API-Key: tsk_your_key" \
  -H "Content-Type: application/json" \
  -d '{
    "agent_id": "agent-appointment-reminder",
    "recipients": [
      {"recipient_phone_number": "+919876543210", "user_data": {"name": "Ravi", "date": "Jan 28"}},
      {"recipient_phone_number": "+919876543211", "user_data": {"name": "Priya", "date": "Jan 29"}}
    ],
    "retry_config": {"enabled": true, "max_retries": 2, "retry_intervals_minutes": [30, 60]}
  }'
```

```
# 2. Schedule
curl -X POST https://your-domain:8005/v1/batches/{batch_id}/schedule \
  -H "X-API-Key: tsk_your_key" \
  -H "Content-Type: application/json" \
  -d '{"scheduled_at": "2025-01-28T09:00:00Z"}'
```

13. Analytics

Retrieve call performance metrics for your tenant. Requires `X-API-Key` . All endpoints support `date_from` and `date_to` query parameters.

METHOD	PATH	DESCRIPTION
GET	/v1/analytics/overview	Aggregate stats: volume, duration, cost, outcomes
GET	/v1/analytics/by-agent	Stats broken down by agent
GET	/v1/analytics/by-batch	Stats broken down by batch
GET	/v1/analytics/timeseries	Daily time-series data for charting

Response — GET /v1/analytics/overview

```
{
  "period": {"date_from": "2025-01-01", "date_to": "2025-01-31"},
  "call_volume": {
    "total": 1200, "connected": 890, "not_connected": 280,
    "pending": 30, "connection_rate": 0.742
  },
  "duration": {
    "total_seconds": 40200.5, "avg_seconds": 45.2,
    "p50_seconds": 38.0, "p90_seconds": 72.0
  },
  "cost": {
    "total": 144.60, "avg_per_call": 0.121, "avg_per_connected": 0.163
  },
  "outcomes": {
    "analyzed_count": 850, "coverage_pct": 95.5,
    "booking": {"count": 340, "pct_of_analyzed": 40.0},
    "escalation": {"count": 85, "pct_of_analyzed": 10.0},
    "not_interested": {"count": 170, "pct_of_analyzed": 20.0},
    "follow_up": {"count": 128, "pct_of_analyzed": 15.1},
    "no_output": {"count": 42, "pct_of_analyzed": 4.9},
    "not_reached": {"count": 60, "pct_of_analyzed": 7.1},
    "other": {"count": 25, "pct_of_analyzed": 2.9}
  },
  "not_connected_breakdown": {"busy": 45, "no_answer": 180, "failed": 55},
  "retry_stats": {"calls_with_retry": 120, "avg_retries": 1.4}
}
```

14. Agents

List and inspect voice agents assigned to your tenant. Requires `X-API-Key`.

METHOD	PATH	DESCRIPTION
GET	/v1/agents	List agents enabled for your tenant
GET	/v1/agents/drift	List drift events (agent changes detected)
GET	/v1/agents/{agent_id}/variables	Get required/optional prompt variables for an agent

Agent Variables

Agents use prompt variables that are injected at call time. Use the variables endpoint to discover what each agent expects:

```
{
  "agent_id": "agent-appointment-reminder",
  "required": ["name", "date", "time", "doctor_name"],
  "optional": ["clinic_address"],
  "system_injected": ["contact_number", "from_number"],
  "all_prompt_variables": ["name", "date", "time", "doctor_name", "clinic_address", "contact_number"]
}
```

Variable Validation: When creating a batch, each recipient's `user_data` must include all `required` variables for the selected agent. Missing variables will cause a validation error.

15. Phone Numbers

List phone numbers assigned to your tenant. Requires `X-API-Key` .

METHOD	PATH	DESCRIPTION
GET	/v1/phone-numbers	List assigned phone numbers and the default caller ID

Response

```
{
  "default_from_number": "+911234567890",
  "phone_numbers": [
    {
      "id": "pn_001",
      "phone_number": "+911234567890",
      "telephony_provider": "twilio",
      "rented": true,
      "renewal_at": "2025-03-01"
    }
  ]
}
```

16. SQL Builder Agent

Status: Disabled — Not Currently Available. The SQL Builder Agent is paused and its endpoint is not mounted in the running application. The feature requires a dedicated read-only PostgreSQL connection (`KALAAM_READONLY_DATABASE_URL`) and LLM API keys that are not configured in the current deployment. The router is gracefully skipped at startup when dependencies are unavailable. This section is retained for future reference.

Natural language to SQL query interface. Ask questions about your call data in plain English. Requires `X-API-Key` and a configured Kalaam read-only database connection.

METHOD	PATH	STATUS	DESCRIPTION
POST	/v1/sql-agent/query	Disabled	Submit a natural language query

Prerequisites (when re-enabled)

The SQL Builder Agent requires the following environment variables to be configured before it can be activated:

VARIABLE	DESCRIPTION
KALAAM_READONLY_DATABASE_URL	Read-only PostgreSQL connection to the target Kalaam DB (dedicated <code>sql_agent_readonly</code> role)
GOOGLE_API_KEY or ANTHROPIC_API_KEY	LLM API key for SQL generation (default model: <code>gemini-2.5-flash</code>)
SQL_AGENT_CONTROL_DB_URL	Database for SQL agent metadata and workspace catalog

Example (for future reference)

```
# This endpoint is currently disabled. When re-enabled:
curl -X POST https://your-domain:8005/v1/sql-agent/query \
  -H "X-API-Key: ts_k_your_key" \
  -H "Content-Type: application/json" \
  -d '{"question": "How many calls were completed last week with a booking outcome?"}'
```

Safety (when enabled): The SQL agent connects via a PostgreSQL role with `default_transaction_read_only=on` and a configurable `statement_timeout` . All generated

queries are validated with `EXPLAIN` before execution and scoped to the tenant's data.

17. Webhooks

Inbound — Voice Engine Callback

METHOD	PATH	AUTH	DESCRIPTION
POST	/webhooks/voice	IP Allowlist	Receives call status updates from the voice engine

This endpoint is called by the Bolna voice engine to report call status changes, transcripts, and extracted data. Configure `TURING_PUBLIC_URL` so the voice engine knows where to send callbacks.

Outbound — Call Outcome Notifications

When a client configures a `webhook_url` in their client config, Turing sends a POST request for each completed call with outcome data.

Webhook Payload

```
POST https://your-webhook-endpoint.com/callback
Content-Type: application/json
X-Turing-Signature: sha256=abc123...

{
  "call_id": "exec_abc123",
  "agent_id": "agent-001",
  "voice_batch_id": "vb_xyz789",
  "contact_number": "+919876543210",
  "status": "completed",
  "duration": 45.2,
  "outcome": "booking",
  "summary": "Patient confirmed appointment"
}
```

Signature Verification

If `webhook_secret` is set, Turing signs the request body with HMAC-SHA256. Verify using:

```
import hmac, hashlib

def verify_signature(body: bytes, secret: str, signature: str) -> bool:
    expected = "sha256=" + hmac.new(
        secret.encode(), body, hashlib.sha256
```



```
).hexdigest()  
return hmac.compare_digest(expected, signature)
```

18. Schema Reference

Complete reference for all request/response models used across the API.

Call Schemas

CallListItem		
FIELD	TYPE	DESCRIPTION
call_id	string null	Voice engine execution ID
agent_id	string	Agent that handled the call
batch_id	uuid null	Batch this call belongs to
contact_number	string null	Recipient phone number
from_number	string null	Caller ID used
status	string	Call status
duration	float null	Duration in seconds
cost	float null	Cost in USD
hangup_reason	string null	Why the call ended
recording_url	string null	URL to call recording
created_at	string null	ISO 8601 timestamp
analysis	CallAnalysisResult null	LLM analysis results

CallDetail (extends CallListItem)		
FIELD	TYPE	DESCRIPTION
transcript	string null	Full conversation transcript
extracted_data	object null	Structured data extracted by the agent
patient_ref	string null	Patient reference identifier

FIELD	TYPE	DESCRIPTION
retry_count	integer null	Number of retry attempts

CallAnalysisResult

FIELD	TYPE	DESCRIPTION
outcome	string	Classification (booking, escalation, etc.)
summary	string null	Brief summary of the call
reason	string null	Explanation for the classification
requests	array[string]	Specific requests made during the call
urgency	string null	Urgency level of the call
confidence	float null	Model confidence (0–1)
symptoms_reported	array[string] null	Health symptoms mentioned
model_used	string null	LLM model used for analysis
analyzed_at	datetime null	When analysis was performed

Admin Schemas

ClientSummary

FIELD	TYPE	DESCRIPTION
id	uuid	Client ID
name	string	Organization name
slug	string	URL-safe identifier
contact_email	string null	Contact email
status	string	pending active suspended rejected
created_at	datetime	Registration timestamp
approved_at	datetime null	Approval timestamp

ApproveResponse

FIELD	TYPE	DESCRIPTION
<code>client_id</code>	<code>uuid</code>	Approved client ID
<code>status</code>	<code>string</code>	New status (active)
<code>api_key</code>	<code>string</code>	Issued API key (shown only once)
<code>claim_url</code>	<code>string null</code>	One-time claim URL (if claim links enabled)

KeySummary

FIELD	TYPE	DESCRIPTION
<code>id</code>	<code>uuid</code>	Key ID
<code>key_prefix</code>	<code>string</code>	First characters for identification
<code>label</code>	<code>string null</code>	Optional label
<code>status</code>	<code>string</code>	active revoked
<code>last_used_at</code>	<code>datetime null</code>	Last API call timestamp
<code>expires_at</code>	<code>datetime null</code>	Expiry (null = no expiry)
<code>created_at</code>	<code>datetime</code>	Issuance timestamp

Batch Schemas**BatchSummary**

FIELD	TYPE	DESCRIPTION
<code>batch_id</code>	<code>string null</code>	Internal batch ID
<code>internal_id</code>	<code>string null</code>	Voice engine batch ID
<code>status</code>	<code>string null</code>	created scheduled in_progress completed stopped
<code>scheduled_at</code>	<code>string null</code>	Scheduled execution time

FIELD	TYPE	DESCRIPTION
file_name	string null	CSV filename (if uploaded)
valid_contacts	integer null	Valid recipients after validation
total_contacts	integer null	Total recipients submitted
from_phone_numbers	array[string] null	Caller IDs used
execution_status	object null	Execution progress counts
created_at	string null	Creation timestamp

BatchMetricsResponse

FIELD	TYPE	DESCRIPTION
batch_id	string	Internal batch ID
voice_batch_id	string null	Voice engine batch ID
status	string	Current batch status
total_recipients	integer	Total recipients
calls_tracked	integer	Calls with tracking data
by_status	object	Call count by status
completed	integer	Successfully completed calls
terminal	integer	Calls in terminal state
success_rate	float null	Completion rate (0–1)
total_cost	float	Total cost in USD
avg_duration_seconds	float null	Average call duration

19. Environment Variables

Complete list of environment variables for deployment configuration.

Core Application

VARIABLE	DEFAULT	DESCRIPTION
APP_NAME	turing_service	Application name
ENVIRONMENT	development	Runtime environment
LOG_LEVEL	INFO	Logging verbosity
HOST	0.0.0.0	Bind address
PORT	8005	Bind port

Database & Cache

VARIABLE	DEFAULT	DESCRIPTION
DATABASE_URL	postgresql+asyncpg://turing:turing@localhost:5433/turing_db	Primary database connection string
REDIS_URL	null	Redis connection (for rate limiting claim limit)

Authentication

VARIABLE	DEFAULT	DESCRIPTION
ADMIN_API_KEY	—	required Shared secret for admin endpoints
API_KEY_CACHE_TTL_SECONDS	60	Tenant API key cache duration
ENCRYPTION_KEY	null	Fernet key for encrypting sensitive config (LLM API keys)

Voice Engine

VARIABLE	DEFAULT	DESCRIPTION
VOICE_ENGINE_API_KEY	—	required Bolna AI API key
VOICE_ENGINE_BASE_URL	https://api.bolna.ai	Voice engine base URL
VOICE_ENGINE_TIMEOUT_SECONDS	30	HTTP timeout for voice engine calls
VOICE_DEFAULT_FROM_NUMBER	null	System-wide default caller ID
VOICE_WEBHOOK_ALLOWED_IPS	""	Comma-separated IPs allowed for voice webhooks
TURING_PUBLIC_URL	null	Public URL for voice engine to send callbacks to

LLM / Analysis

VARIABLE	DEFAULT	DESCRIPTION
LLM_PROVIDER	anthropic	Default LLM provider for call analysis
LLM_MODEL	null	Model override (provider default if null)
ANTHROPIC_API_KEY	null	Anthropic API key
OPENAI_API_KEY	null	OpenAI API key
GOOGLE_API_KEY	null	Google API key (for SQL Agent)

Features & Limits

VARIABLE	DEFAULT	DESCRIPTION
REGISTER_RATE_LIMIT_PER_HOUR	10	Max registration attempts per IP per hour
CLAIM_LINK_TTL_HOURS	24	Claim link expiry in hours
AGENT_SYNC_INTERVAL_MINUTES	15	Auto-sync interval for agent catalog
CONSOLE_PUBLIC_URL	null	Frontend console URL (for claim link generation)

SQL Agent (Advanced)

VARIABLE	DEFAULT	DESCRIPTION
SQL_AGENT_CONTROL_DB_URL	—	Database for SQL agent metadata
KALAAM_READONLY_DATABASE_URL	—	Read-only connection to Kalaam production DB
SQL_AGENT_EXPLAIN_VALIDATION	true	Validate generated SQL with EXPLAIN
SQL_AGENT_MAX_REPAIR_ATTEMPTS	3	Max LLM repair attempts for invalid SQL

20. Changelog

VERSION	DATE	CHANGES
1.0.0	2026-08-25	Initial integration guide covering all 68 endpoints across 13 routers. Includes: multi-tenant auth, admin console, batch management, analytics, call analysis, SQL builder agent, webhook flows, and full schema reference.